13 The table shows a set of values for x and y.

x	1	2	3	4
y	9	$2\frac{1}{4}$	1	$\frac{9}{16}$

y is inversely proportional to the square of x.

(a) Find an equation for y in terms of x.

Inverse square: $y = \frac{k}{x^2}$

1 mark
Start with the inverse-square model

At $x = 1$ and $y = 9$, $k = yx^2 = 9$.

Therefore $y = \frac{9}{x^2}$

1 mark
Correct equation

$y = \frac{9}{x^2}$ **(2)**

(b) Find the positive value of x when $y = 16$

$16 = \frac{9}{x^2}$ so $16x^2 = 9$. $x^2 = \frac{9}{16}$

1 mark
Substitute y = 16 into the equation

1 mark
Take the positive square root

Positive $x = \frac{3}{4}$ $\frac{3}{4}$ **(2)**

(Total for Question 13 is 4 marks)